Tugas 2
Perceptron

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D4 TEKNIK KOMPUTER
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TUGAS TEORI

Tugas :

1. Buatlah program sederhana untuk kasus sederhana single perceptron AND.
2. Buatlah program sederhana untuk kasus sederhana single perceptron OR.
3. Buat program untuk 7 segment input 7 output 10

1. Perceptron AND

Program:

perceptron_and.cpp > main()

```c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
double d_rand(){
    return(((double)(rand()%32767)/32767.0-0.5)*2.0);
}
int main() {
    int i,j,out,ERR,y;
    int x[3][4]= {
        {0,0,1,1},
        {0,1,0,1},
        {1,1,1,1}
    };
    int T[4]={0,0,0,1};
    float w[3],O,LR=0.1,init=0.15;
    //inisialisasi bobot
    for(i=0;i<3;i++){
        w[i]=init*d_rand();
    }
    //traning
    for(i=0;i<10;i++){
        for(j=0;j<4;j++){
            O=x[0][j]*w[0]+x[1][j]*w[1]+x[2][j]*w[2];
            if(O>0.0)
                out=1;
            else
                out=0;
            ERR=T[j]-out;
            if(ERR!=0){
                w[0]=w[0]+LR*x[0][j]*ERR;
                w[1]=w[1]+LR*x[1][j]*ERR;
                w[2]=w[2]+LR*x[2][j]*ERR;
            }
        }
        printf("Err:%d\n",ERR);
    }
    //running
    for(i=0;i<4;i++){
        printf("Input X1:");
        scanf("%d",&x[0][0]);
        printf("Input X2:");
        scanf("%d",&x[1][0]);
        O=x[0][0]*w[0]+x[1][0]*w[1]+x[2][0]*w[2];
        if(O>0.0)
            y=1;
        else
            y=0;
        printf("Output:%d\n",y);
    }
    return 0;
}
```

Output:

```
Err:1
Err:1
Err:1
Err:1
Err:1
Err:1
Err:0
Err:0
Err:0
Err:0
Input X1:0
Input X2:0
Output:0
Input X1:0
Input X2:1
Output:0
Input X1:1
Input X2:0
Output:0
Input X1:1
Input X2:1
Output:1
```

Truth table AND:

x1	x2	Output
0	0	0
0	1	0
1	0	0
1	1	1

2. Perceptron OR

Program:

perceptron_or.cpp > main()

```cpp
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
double d_rand(){
    return(((double)(rand()%32767)/32767.0-0.5)*2.0);
}
int main() {
    int i,j,out,ERR,y;
    int x[3][4]= {
        {0,0,1,1},
        {0,1,0,1},
        {1,1,1,1}
    };
    int T[4]={0,1,1,1};
    float w[3],O,LR=0.1,init=0.15;
    //inisialisasi bobot
    for(i=0;i<3;i++){
        w[i]=init*d_rand();
    }
    //traning
    for(i=0;i<10;i++){
        for(j=0;j<4;j++){
            O=x[0][j]*w[0]+x[1][j]*w[1]+x[2][j]*w[2];
            if(O>0.0)
                out=1;
            else
                out=0;
            ERR=T[j]-out;
            if(ERR!=0){
                w[0]=w[0]+LR*x[0][j]*ERR;
                w[1]=w[1]+LR*x[1][j]*ERR;
                w[2]=w[2]+LR*x[2][j]*ERR;
            }
        }
        printf("Err:%d\n",ERR);
    }
    //running
    for(i=0;i<4;i++){
        printf("Input X1:");
        scanf("%d",&x[0][0]);
        printf("Input X2:");
        scanf("%d",&x[1][0]);
        O=x[0][0]*w[0]+x[1][0]*w[1]+x[2][0]*w[2];
        if(O>0.0)
            y=1;
        else
            y=0;
        printf("Output:%d\n",y);
    }
    return 0;
}
```

Output:

```
Err:0
Err:0
Err:0
Err:0
Err:0
Err:0
Err:0
Err:0
Err:0
Err:0
Input X1:0
Input X2:0
Output:0
Input X1:0
Input X2:1
Output:1
Input X1:1
Input X2:0
Output:1
Input X1:1
Input X2:1
Output:1
```

Truth table OR:

x1	x2	Output
0	0	0
0	1	1
1	0	1
1	1	1

3. Perceptron OR
   Program:

perceptron_7segment.cpp > ...

```cpp
#include <stdio.h>
#include <stdlib.h>
double d_rand() {
    return (((double)(rand() % 32767) / 32767.0 - 0.5) * 2.0);
}
int main() {
    int i, j, k, out, ERR, y;
    int x[10][7] = {
        {1, 1, 1, 1, 1, 1, 0}, // 0
        {0, 1, 1, 0, 0, 0, 0}, // 1
        {1, 1, 0, 1, 1, 0, 1}, // 2
        {1, 1, 1, 1, 0, 0, 1}, // 3
        {0, 1, 1, 0, 0, 1, 1}, // 4
        {1, 0, 1, 1, 0, 1, 1}, // 5
        {1, 0, 1, 1, 1, 1, 1}, // 6
        {1, 1, 1, 0, 0, 0, 0}, // 7
        {1, 1, 1, 1, 1, 1, 1}, // 8
        {1, 1, 1, 1, 0, 1, 1}};  // 9
    int T[10][10] = {
        {1, 0, 0, 0, 0, 0, 0, 0, 0, 0}, // 0
        {0, 1, 0, 0, 0, 0, 0, 0, 0, 0}, // 1
        {0, 0, 1, 0, 0, 0, 0, 0, 0, 0}, // 2
        {0, 0, 0, 1, 0, 0, 0, 0, 0, 0}, // 3
        {0, 0, 0, 0, 1, 0, 0, 0, 0, 0}, // 4
        {0, 0, 0, 0, 0, 1, 0, 0, 0, 0}, // 5
        {0, 0, 0, 0, 0, 0, 1, 0, 0, 0}, // 6
        {0, 0, 0, 0, 0, 0, 0, 1, 0, 0}, // 7
        {0, 0, 0, 0, 0, 0, 0, 0, 1, 0}, // 8
        {0, 0, 0, 0, 0, 0, 0, 0, 0, 1}};  // 9
    float w[7][10], O[10], LR = 0.1, init = 0.15;
    for (i = 0; i < 7; i++) {
        for (j = 0; j < 10; j++) {
            w[i][j] = init * d_rand();
        }
    }
    int count = 0;
    for (k = 0; k < 50; k++) {
        for (i = 0; i < 10; i++) {
            for (j = 0; j < 10; j++) {
                O[j] = 0;
                for (int l = 0; l < 7; l++) {
                    O[j] += x[i][l] * w[l][j];
                }
                out = (O[j] > 0.0) ? 1 : 0;
                ERR = T[i][j] - out;
                if (ERR != 0) {
                    for (int l = 0; l < 7; l++) {
                        w[l][j] += LR * x[i][l] * ERR;
                    }}}}
        printf("Epoch %d ERR = %d\n", count, ERR); // Hanya mengubah printf
        count++;}
    int input[7];
    printf("Masukkan input 7-segment (7 angka 0 atau 1):\n");
    for (i = 0; i < 7; i++) {
        scanf("%d", &input[i]);
    }
    for (j = 0; j < 10; j++) {
        O[j] = 0;
        for (i = 0; i < 7; i++) {
            O[j] += input[i] * w[i][j];
        }
        out = (O[j] > 0.0) ? 1 : 0;
        if (out == 1) {
            printf("Output: %d\n", j);
            break;
        }
    }
    return 0;
}
```

Output:

```
Epoch 1 ERR = 1
Epoch 2 ERR = 1
Epoch 3 ERR = 1
Epoch 4 ERR = 1
Epoch 5 ERR = 1
Epoch 6 ERR = 1
Epoch 7 ERR = 1
Epoch 8 ERR = 1
Epoch 9 ERR = 1
Epoch 10 ERR = 1
Epoch 11 ERR = 1
Epoch 12 ERR = 1
Epoch 13 ERR = 1
Epoch 14 ERR = 1
Epoch 15 ERR = 1
Epoch 16 ERR = 1
Epoch 17 ERR = 1
Epoch 18 ERR = 1
Epoch 19 ERR = 1
Epoch 20 ERR = 1
Epoch 21 ERR = 1
Epoch 22 ERR = 1
Epoch 23 ERR = 1
Epoch 24 ERR = 1
Epoch 25 ERR = 1
Epoch 26 ERR = 1
Epoch 27 ERR = 1
Epoch 28 ERR = 0
Epoch 29 ERR = 0
Epoch 30 ERR = 0
Epoch 31 ERR = 0
Epoch 32 ERR = 0
Epoch 33 ERR = 0
Epoch 34 ERR = 0
Epoch 35 ERR = 0
Epoch 36 ERR = 0
Epoch 37 ERR = 0
Epoch 38 ERR = 0
Epoch 39 ERR = 0
Epoch 40 ERR = 0
Epoch 41 ERR = 0
Epoch 42 ERR = 0
Epoch 43 ERR = 0
Epoch 44 ERR = 0
Epoch 45 ERR = 0
Epoch 46 ERR = 0
Epoch 47 ERR = 0
Epoch 48 ERR = 0
Epoch 49 ERR = 0
Masukkan input 7-segment (7 angka 0 atau 1):
```

```
Masukkan input 7-segment (7 angka 0 atau 1):
1
1
1
1
1
1
0
Output: 0
Masukkan input 7-segment (7 angka 0 atau 1):
0
1
1
0
0
0
0
Output: 1
Masukkan input 7-segment (7 angka 0 atau 1):
1
1
0
1
1
0
1
Output: 2
Masukkan input 7-segment (7 angka 0 atau 1):
1
1
1
1
0
0
1
Output: 3
Masukkan input 7-segment (7 angka 0 atau 1):
0
1
1
0
0
1
1
Output: 4
Masukkan input 7-segment (7 angka 0 atau 1):
1
0
1
1
0
1
1
Output: 5
```

```
Masukkan input 7-segment (7 angka 0 atau 1):
1
1
1
0
0
0
0
Output: 7
Masukkan input 7-segment (7 angka 0 atau 1):
1
1
1
1
1
1
1
Output: 8
Masukkan input 7-segment (7 angka 0 atau 1):
1
1
1
1
0
1
1
Output: 9
```

Truth table Seven Segment

A	B	C	D	E	F	G	Display
1	1	1	1	1	1	0	0
0	1	1	0	0	0	0	1
1	1	0	1	1	0	1	2
1	1	1	1	0	0	1	3
0	1	1	0	0	1	1	4
1	0	1	1	0	1	1	5
1	0	1	1	1	1	1	6
1	1	1	0	0	0	0	7
1	1	1	1	1	1	1	8
1	1	1	1	0	1	1	9